The Method of Moments

The method of moments consists of substituting sample moments such as the sample mean, $\bar{y}$, sample variance $\hat{\gamma}_0$ and sample ACF $\hat{\rho}_i$ for their theoretical counterparts and solving the resultant equation to obtain estimates of unknown parameters. For example, in the AR (P) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + z_t$$

$$E(y_{t-k} y_t) = \phi_1 E(y_{t-k} y_{t-1}) + \phi_2 E(y_{t-k} y_{t-2}) + \cdots + \phi_p E(y_{t-k} y_{t-p}) + E(y_{t-k} z_t).$$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \cdots + \phi_p \gamma_{k-p} \qquad k>0 \text{ or } k \geq 1$$

Note that $E[y_{t-k} z_t] = 0$ for $k>0$ since y_{t-k} can only involves the shocks z_j up to time $t-k$ which are uncorrelated with z_t.

$E(z_{t-0} z_t) = E(z_t^2) = \sigma_z^2$

$$y_{t-k} = \phi_1 y_{t-k-1} + \phi_2 y_{t-k-2} + \cdots + \phi_p y_{t-k-p} + z_{t-k}$$

Now divide both sides by γ_0

$$\rho_K = \phi_1 \rho_{K-1} + \cdots + \phi_p \rho_{K-p}, \quad K > 0 \text{ or } K \geq 1$$

To estimate ϕ, we use above equation to obtain the following system of Yule-Walker equations:

$$K=1;\quad \rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1} + \phi_3 \rho_{-2} + \cdots + \phi_p \rho_{1-p}$$

$$K=2;\quad \rho_2 = \phi_1 \rho_1 + \phi_2 \rho_0 + \phi_3 \rho_{-1} + \cdots + \phi_p \rho_{2-p}$$

$$\vdots \quad \vdots$$

$$K=p,\quad \rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \phi_3 \rho_{p-3} + \cdots + \phi_p \rho_0$$

since $\boxed{\rho_0 = 1}$ $\boxed{\rho_K = \rho_{-K}}$

$$\rho_1 = \phi_1 + \phi_2 \rho_1 + \phi_3 \rho_2 + \cdots + \phi_p \rho_{p-1}$$

$$\rho_2 = \phi_1 \rho_1 + \phi_2 + \phi_3 \rho_1 + \cdots + \phi_p \rho_{p-2}$$

$$\vdots$$

$$\rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \phi_3 \rho_{p-2} + \cdots + \phi_p$$

Replace ρ_K by r_K, we obtain the moment estimators $\hat{\phi}_1, \hat{\phi}_2, \ldots, \hat{\phi}_p$ by solving linear system of equations

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \vdots \\ \hat{\phi}_p \end{bmatrix} = \begin{bmatrix} 1 & r_1 & r_2 & \cdots & r_{p-2} & r_{p-1} \\ r_1 & 1 & r_1 & \cdots & r_{p-3} & r_{p-2} \\ \vdots & & & & & \\ r_{p-1} & r_{p-2} & r_{p-3} & \cdots & r_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_p \end{bmatrix}$$

These estimators are usually called "Yule-Walker estimators".

Having obtained $\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_p$ we use the result

$$\gamma_0 = E(y_t y_t) = E\left[y_t(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + z_t)\right]$$

$$\gamma_0 = \phi_1 E(y_t y_{t-1}) + \phi_2 E(y_t y_{t-2}) + \dots + \phi_p E(y_t y_{t-p}) + E(y_t z_t)$$

$$\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \dots + \phi_p \gamma_p + \sigma_z^2$$

$$\sigma_z^2 = \gamma_0 - \phi_1 \gamma_1 - \phi_2 \gamma_2 - \dots - \phi_p \gamma_p$$

$$\sigma_z^2 / \gamma_0 = 1 - \phi_1 \rho_1 - \phi_2 \rho_2 - \dots - \phi_p \rho_p$$

$$\sigma_z^2 = \gamma_0 (1 - \phi_1 \rho_1 - \phi_2 \rho_2 - \dots - \phi_p \rho_p)$$

The moment estimator for σ_z^2

$$\hat{\sigma}_z^2 = \hat{\gamma}_0 (1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2 - \dots - \hat{\phi} r_p)$$

Example: For the AR(1) process

$$y_t - \mu = \phi_1 (y_{t-1} - \mu) + z_t$$

the Yule-Walker estimator for ϕ_1 is

$$\hat{\phi}_1 = r_1$$

The moment estimator for μ and σ_z^2 are

$$\hat{\mu} = \bar{y}$$

Since $\hat{\sigma}_z^2 = \hat{\gamma}_0 (1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2 - \cdots - \hat{\phi}_p r_p)$

$$\hat{\sigma}_z^2 = \hat{\gamma}_0 (1 - \hat{\phi}_1 r_1),$$

where $\hat{\gamma}_0$ is the sample variance of the y_t series.

Yule-Walker estimation of autoregressive (AR) process

Consider a zero mean AR(p) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + z_t$$

Yule-walker equations: for $k > 0$, we have recursive rule

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \cdots + \phi_p \rho_{k-p}, \text{ for } k > 0$$

put $k = 1, 2, \ldots, p$ in the recursive equation, we obtain

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1} + \cdots + \phi_p \rho_{1-p}$$
$$\rho_2 = \phi_1 \rho_1 + \phi_2 \rho_0 + \cdots + \phi_p \rho_{2-p}$$
$$\vdots$$
$$\rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \cdots + \phi_p \rho_0$$

system of Yule-Walker equations

we know that $\rho_0 = 1$ and $\rho_{-k} = \rho_k$, we can write the above 'p' equations as

$$\rho_1 = \phi_1 + \phi_2 \rho_1 + \cdots + \phi_p \rho_{1-p}$$
$$\rho_2 = \phi_1 \rho_1 + \phi_2 + \cdots + \phi_p \rho_{2-p}$$
$$\vdots$$
$$\rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \cdots + \phi_p$$

which are 'p' equations which relates 'p' parameters of AR(p) with the Autocorrelation of the process.

In matrix form

$$\begin{bmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_p \end{bmatrix} = \begin{bmatrix} 1 & \rho_1 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & \cdots & \rho_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \cdots & 1 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{bmatrix}$$

$$\underline{\rho} = \underset{\sim}{\rho}\, \underline{\phi} \Rightarrow \underline{\phi} = \underset{\sim}{\rho}^{-1} \underline{\rho}$$

using the theory of method of moments estimation

$$r_K = \hat{\phi}_1 r_{K-1} + \hat{\phi}_2 r_{K-2} + \cdots + \hat{\phi}_p r_{K-p}$$

put $K = 1, 2, \cdots, P$ and using $r_0 = 1$ and $r_{-K} = r_K$, we obtain

$$r_1 = \hat{\phi}_1 + \hat{\phi}_2 r_1 + \cdots + \hat{\phi}_p r_{p-1}$$
$$r_2 = \hat{\phi}_1 r_1 + \hat{\phi}_2 + \cdots + \hat{\phi}_p r_{p-2}$$
$$\vdots$$
$$r_p = \hat{\phi}_1 r_{p-1} + \hat{\phi}_2 r_{p-2} + \cdots + \hat{\phi}_p$$

In matrix form:

$$\begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_p \end{bmatrix} = \begin{bmatrix} 1 & r_1 & \cdots & r_{p-1} \\ r_1 & 1 & \cdots & r_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p-1} & r_{p-2} & \cdots & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \vdots \\ \hat{\phi}_p \end{bmatrix}$$

These "estimators" are usually called "Yule-Walker estimators".

$$\underline{r} = \underset{\sim}{R}\,\underline{\hat{\phi}} \Rightarrow \underline{\hat{\phi}} = \underset{\sim}{R}^{-1}\underline{r}$$

Yule-walker estimation of AR(1)

$p = 1$, $\quad y_t = \phi_1 y_{t-1} + z_t \longrightarrow AR(1)$

$$\rho_K = \phi_1 \rho_{K-1}$$

$$\rho_1 = \phi_1 \rho_0 \quad \text{for } K = 1$$

$$\phi_1 = \rho_1 \quad \because \rho_0 = 1$$

$$\boxed{\hat{\phi}_1 = r_1} \rightarrow \text{Yule - walker estimate of } \phi_1$$

The moment estimator for σ_z^2

$$\hat{\sigma}_z^2 = \hat{\gamma}_0 (1 - \hat{\phi}_1 r_1)$$

$$\hat{\sigma}_z^2 = \hat{\gamma}_0 (1 - r_1^2)$$

where $\hat{\gamma}_0$ is the sample variance of the $\{y_t\}_{t=1}^{n}$ series.

Yule- walker estimation of AR (2) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + z_t$$

$$\rho_K = \phi_1 \rho_{K-1} + \phi_2 \rho_{K-2}, \quad \text{for } K \geq 1$$

The Yule - Walker equations are

$K=1: \quad \rho_1 = \phi_1 + \phi_2 \rho_1$

$K=2: \quad \rho_2 = \phi_1 \rho_1 + \phi_2$

Replace the process ACF with the sample ACF, we get

$$r_1 = \hat{\phi}_1 + \hat{\phi}_2 r_1$$

$$r_2 = \hat{\phi}_1 r_1 + \hat{\phi}_2$$

$$\begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} 1 & r_1 \\ r_1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} 1 & r_1 \\ r_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$= \frac{1}{1 - r_1^2} \begin{bmatrix} 1 & -r_1 \\ -r_1 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \frac{1}{1 - r_1^2} \begin{bmatrix} r_1 - r_1 r_2 \\ -r_1^2 + r_2 \end{bmatrix}$$

$$\boxed{\hat{\phi}_1 = \frac{r_1 - r_1 r_2}{1 - r_1^2}}, \quad \boxed{\hat{\phi}_2 = \frac{r_2 - r_1^2}{1 - r_1^2}}$$

①

Estimate the Parameter of AR(3) process by using moment estimation.

AR(3) process: $\dot{z}_t = \phi_1 \dot{z}_{t-1} + \phi_2 \dot{z}_{t-2} + \phi_3 \dot{z}_{t-3} + a_t$

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \phi_3 \rho_{k-3}, \quad k \geq 1$$

System of Yule-Walker equations

$$\rho_1 = \phi_1 + \phi_2 \rho_1 + \phi_3 \rho_2$$
$$\rho_2 = \phi_1 \rho_1 + \phi_2 + \phi_3 \rho_1$$
$$\rho_3 = \phi_1 \rho_2 + \phi_2 \rho_1 + \phi_3$$

Replace ρ_k by r_k, Moment estimators $\hat{\phi}_1, \hat{\phi}_2$ and $\hat{\phi}_3$ can be obtained by solving the above linear system of equations.

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \hat{\phi}_3 \end{bmatrix} = \begin{bmatrix} 1 & r_1 & r_2 \\ r_1 & 1 & r_1 \\ r_2 & r_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \operatorname{Adj}(A)$$

$$|A| = 1(1 - r_1^2) - r_1(r_1 - r_1 r_2) + r_2(r_1^2 - r_2)$$
$$= 1 - r_1^2 - r_1^2 + r_1^2 r_2 + r_1^2 r_2 - r_2^2$$
$$|A| = 1 - 2r_1^2 - r_2^2 + 2r_1^2 r_2$$

First row elements

$$\begin{vmatrix} 1 & \lambda_1 \\ \lambda_1 & 1 \end{vmatrix} = 1-\lambda_1^2, \quad \begin{vmatrix} \lambda_1 & \lambda_1 \\ \lambda_2 & 1 \end{vmatrix} = \lambda_1 - \lambda_1\lambda_2, \quad \begin{vmatrix} \lambda_1 & 1 \\ \lambda_2 & \lambda_1 \end{vmatrix} = \lambda_1^2-\lambda_2$$

Second row elements

$$\begin{vmatrix} \lambda_1 & \lambda_2 \\ \lambda_1 & 1 \end{vmatrix} = \lambda_1 - \lambda_1\lambda_2, \quad \begin{vmatrix} 1 & \lambda_2 \\ \lambda_2 & 1 \end{vmatrix} = 1-\lambda_2^2, \quad \begin{vmatrix} 1 & \lambda_1 \\ \lambda_2 & \lambda_1 \end{vmatrix} = \lambda_1 - \lambda_1\lambda_2$$

3rd row elements

$$\begin{vmatrix} \lambda_1 & \lambda_2 \\ 1 & \lambda_1 \end{vmatrix} = \lambda_1^2-\lambda_2, \quad \begin{vmatrix} 1 & \lambda_2 \\ \lambda_1 & \lambda_1 \end{vmatrix} = \lambda_1 - \lambda_1\lambda_2, \quad \begin{vmatrix} 1 & \lambda_1 \\ \lambda_1 & 1 \end{vmatrix} = 1-\lambda_1^2$$

Matrix of cofactors

$$\begin{bmatrix} 1-\lambda_1^2 & \lambda_1-\lambda_1\lambda_2 & \lambda_1^2-\lambda_2 \\ \lambda_1-\lambda_1\lambda_2 & 1-\lambda_2^2 & \lambda_1-\lambda_1\lambda_2 \\ \lambda_1^2-\lambda_2 & \lambda_1-\lambda_1\lambda_2 & 1-\lambda_1^2 \end{bmatrix} \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Hence

$$Adj(A) = \begin{bmatrix} 1-\lambda_1^2 & \lambda_1\lambda_2-\lambda_1 & \lambda_1^2-\lambda_2 \\ \lambda_1\lambda_2-\lambda_1 & 1-\lambda_2^2 & \lambda_1\lambda_2-\lambda_1 \\ \lambda_1^2-\lambda_2 & \lambda_1\lambda_2-\lambda_1 & 1-\lambda_1^2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1-2\lambda_1^2-\lambda_2^2+2\lambda_1^2\lambda_2} \begin{bmatrix} 1-\lambda_1^2 & \lambda_1\lambda_2-\lambda_1 & \lambda_1^2-\lambda_2 \\ \lambda_1\lambda_2-\lambda_1 & 1-\lambda_2^2 & \lambda_1\lambda_2-\lambda_1 \\ \lambda_1^2-\lambda_2 & \lambda_1\lambda_2-\lambda_1 & 1-\lambda_1^2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \hat{\phi}_3 \end{bmatrix} = \frac{1}{1-2\rho_1^2-\rho_2^2+2\rho_1^2\rho_2} \begin{bmatrix} 1-\rho_1^2 & \rho_1\rho_2-\rho_1 & \rho_1^2-\rho_2 \\ \rho_1\rho_2-\rho_1 & 1-\rho_2^2 & \rho_1\rho_2-\rho_1 \\ \rho_1^2-\rho_2 & \rho_1\rho_2-\rho_1 & 1-\rho_1^2 \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix}$$

$$\hat{\phi}_1 = \frac{\rho_1(1-\rho_1^2) + \rho_2(\rho_1\rho_2-\rho_1) + \rho_3(\rho_1^2-\rho_2)}{1-2\rho_1^2-\rho_2^2+2\rho_1^2\rho_2}$$

$$\hat{\phi}_2 = \frac{\rho_1(\rho_1\rho_2-\rho_1) + \rho_2(1-\rho_2^2) + \rho_3(\rho_1\rho_2-\rho_1)}{1-2\rho_1^2-\rho_2^2+2\rho_1^2\rho_2}$$

$$\hat{\phi}_3 = \frac{\rho_1(\rho_1^2-\rho_2) + \rho_2(\rho_1\rho_2-\rho_1) + \rho_3(1-\rho_1^2)}{1-2\rho_1^2-\rho_2^2+2\rho_1^2\rho_2}$$

$$\rho_1 = \hat{\phi}_1 + \hat{\phi}_2$$

Consider MA(1) process

$$Y_t = Z_t - \theta_1 Z_{t-1}$$

For θ_1 we use that

$$\rho_K = \begin{cases} \dfrac{-\theta_1}{1+\theta_1^2} & ; K=1 \\ 0 & ; K>1 \end{cases}$$

$$\rho_1 = \frac{-\theta_1}{1+\theta_1^2}$$

$$\rho_1 + \rho_1\theta_1^2 = -\theta_1$$

$$\rho_1\theta_1^2 + \theta_1 + \rho_1 = 0$$

$ax^2 + bx + c = 0$

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\boxed{a = \rho_1 \qquad b = 1 \qquad c = \rho_1}$$

$$\theta_1 = \frac{-1 \pm \sqrt{1 - 4\rho_1^2}}{2\rho_1}$$

Replace ρ_1 by $\hat{\rho}_1$

$$\hat{\theta}_1 = \frac{-1 \pm \sqrt{1 - 4\hat{\rho}_1^2}}{2\hat{\rho}_1}$$

$$\boxed{\begin{array}{l} \hat{\theta}_1 = \dfrac{-1 \pm \sqrt{1 - 4\cdot\frac{1}{4}}}{2\cdot\frac{1}{2}} = \\ \hat{\theta}_1 = \pm 1 \end{array}}$$

If $\hat{\rho}_1 = \pm 0.5$, then we have unique solution $\hat{\theta}_1 = \pm 1$ which gives a non invertible estimated model.

If $|\hat{\rho}_1| > 0.5$, then real-valued moment estimator $\hat{\theta}_1$ does not exist.

If $|\hat{\rho}_1| < 0.5$, Then there exist two distinct real-valued solutions and we always choose the one that satisfy the invertible Condition.

$$\hat{\sigma}_a^2 = \hat{\gamma}_0 / 1 + \hat{\theta}_1^2$$

- Estimators for MA and mixed ARMA models are complicated.
- The moment estimates are very sensitive to rounding errors.
- They are usually used to provide intial estimates needed for a more efficient nonlinear estimation, particularly for the MA and mixed ARMA models.
- The moment estimators are not recommended for final estimation results.
- The moment estimators should not be used if The process is close to being nonstationary or noninvertible.

Q.5 (b) For AR(3) process, show that

$$\rho_1 = \frac{\phi_1 + \phi_2\phi_3}{1 - \phi_2 - \phi_3^2 - \phi_1\phi_3}$$

by using recursive rule.

Sol.

AR(3) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + z_t$$

$$\rho_K = \phi_1 \rho_{K-1} + \phi_2 \rho_{K-2} + \phi_3 \rho_{K-3}, \quad K > 0 \quad (1)$$

Substitute $K = 1, 2, 3$ in equation (1), we obtain a set of linear equations for ρ_1, ρ_2 and ρ_3 in terms of ϕ_1, ϕ_2 and ϕ_3, i.e.,

$$\rho_1 = \phi_1 + \phi_2\rho_1 + \phi_3\rho_2 \longrightarrow (2)$$

$$\rho_2 = \phi_1\rho_1 + \phi_2 + \phi_3\rho_1 \longrightarrow (3)$$

$$\rho_3 = \phi_1\rho_2 + \phi_2\rho_1 + \phi_3 \longrightarrow (4)$$

These are the well-known Yule-Walker equations.

Now substitute (3) in (2), we get

Q5 (b)

$$\rho_1 = \phi_1 + \phi_2 \rho_1 + \phi_3 [\phi_1 \rho_1 + \phi_2 + \phi_3 \rho_1]$$

$$\rho_1 = \phi_1 + \phi_2 \rho_1 + \phi_1 \phi_3 \rho_1 + \phi_2 \phi_3 + \phi_3^2 \rho_1$$

$$\rho_1 - \phi_2 \rho_1 - \phi_1 \phi_3 \rho_1 - \phi_3^2 \rho_1 = \phi_1 + \phi_2 \phi_3$$

$$\rho_1 [1 - \phi_2 - \phi_3^2 - \phi_1 \phi_3] = \phi_1 + \phi_2 \phi_3$$

$$\rho_1 = \frac{\phi_1 + \phi_2 \phi_3}{1 - \phi_2 - \phi_3^2 - \phi_1 \phi_3}$$